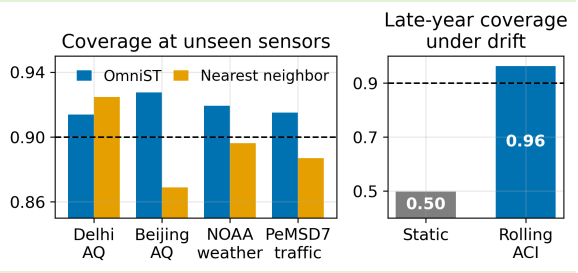
OmniST: Topology-Invariant Forecasting for Dynamic Sensor Networks With Guaranteed Coverage at Unseen Sensors

Praveg Vashishtha

***Abstract*— Deployed sensor networks change: stations are added, fail, or go offline for maintenance. Graph convolutional forecasters built on a fixed adjacency matrix tie their weights to the sensors seen in training, so they cannot forecast at a new location and they fail when sensors are removed. We present OmniST, a forecaster with no structural dependency on the number of sensors. Each sensor is embedded from its geographic coordinates, spatial context comes from nearest-neighbor attention recomputed at every forward pass, and each sensor's history is processed independently. Combined with inductive split-conformal calibration, OmniST returns distribution-free prediction intervals with a finite-sample coverage guarantee at sensor locations never seen in training. We show that a sensor's nonconformity score is unchanged by topology changes outside its neighborhood, so in that regime the guarantee needs no recalibration. On four real sensor datasets covering air quality, weather and freeway traffic, coverage at unseen sensors stays at or above 90% under 30 to 50% sensor dropout, while four interpolation baselines each fall short on several datasets, and it holds at 325 real traffic sensors. At unseen air-quality stations, point error is 12% below nearest-neighbor interpolation and 21% below a learned kriging network. A rolling adaptive-conformal layer restores coverage under seasonal drift, lifting late-year coverage from 0.50 to 0.96.**



***Index Terms*— Conformal prediction, inductive prediction, sensor failure, sensor networks, spatiotemporal forecasting, uncertainty quantification.**

I. Introduction

An operational sensor network rarely keeps the layout it was trained on. India's Central Pollution Control Board adds air-quality stations every year [1], traffic detectors fail, and weather stations go offline for maintenance. A forecaster serving such a network has to keep working when a sensor disappears, and it has to say something useful at a location where a new sensor is planned or has just been installed.

Spatiotemporal graph convolutional networks (STGCNs) [2] forecast accurately on a fixed network, but they encode it as a pre-computed $N \times N$ adjacency matrix. A sensor added after training has no row or column in that matrix, so the model produces no forecast there, and removing a trained sensor breaks the matrix product. Some implementations recompute the adjacency and can adapt, but that is an engineering choice rather than something the formulation provides.

A point forecast at a new location is also not enough for an operator. Before acting on it, they need to know how far to trust it. This paper asks whether a forecaster can deliver both: forecasts at any set of sensors, including unseen ones, and intervals at those sensors that carry a finite-sample coverage guarantee which survives when the network changes.

Our answer is OmniST, a design in which no component references a global sensor count. We use "topology invariant" in this sense: the model produces valid outputs for any number of sensors. We do not claim that accuracy is invariant, though it is stable in practice. The contributions are:

1) **Guaranteed-coverage intervals at unseen sensors.** Composing coordinate-based inductive prediction with inductive split-conformal calibration [3] yields distribution-free intervals at sensor locations with no training history. We prove a locality property (Proposition 1): each sensor's nonconformity score is exactly invariant to sensors outside its neighborhood, so the guarantee needs no recalibration under far-field topology changes.
2) **Resilience to sensor failure.** Accuracy degrades by at most 1.1% when 40% of stations are removed at inference, and coverage holds under 30 to 50% dropout, while a fixed-topology baseline fails outright.
3) **Multi-domain validation on real sensor data.** Coverage at unseen sensors holds on four real datasets

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from three physical domains (Delhi and Beijing air quality, NOAA weather, Caltrans traffic) and at 325 real traffic sensors. Among OmniST and four model-free interpolators, only OmniST holds the target on every dataset.

4) **Coverage over time.** A rolling-window quantile combined with Adaptive Conformal Inference [4] restores coverage under seasonal drift, where static conformal fails.

II. Method

A. An Architecture With No Global Sensor Count

OmniST satisfies one invariant: for any component f and any sensor i, $f(i)$ depends only on sensor i's own data and a fixed-size neighborhood $\mathcal{N}_k(i)$, never on the full set of sensors present.

Coordinate tokens. Each observation becomes a token

$$\mathbf{z}_i(t) = f_{\text{MLP}}(\phi_i, \lambda_i) + f_{\text{time}}(t) + W_f \mathbf{x}_i(t), \tag{1}$$

where $f_{\text{MLP}} : \mathbb{R}^2 \to \mathbb{R}^d$ is a 3-layer MLP on centered latitude and longitude, f_{time} encodes calendar features with sinusoidal harmonics, and W_f projects the raw measurements. No other sensor enters $\mathbf{z}_i(t)$.

Neighbor attention. Spatial context comes from the k geographically nearest sensors,

$$\mathbf{z}'_i = \text{LayerNorm}\Big(\mathbf{z}_i + \sum_{j \in \mathcal{N}_k(i)} \alpha_{ij} W_v \mathbf{z}_j\Big), \tag{2}$$

where $\mathcal{N}_k(i)$ is found by Haversine distance at every forward pass and α_{ij} are multi-head attention weights with a learned distance bias. When sensors are removed, the remaining ones find their neighbors among those still reporting.

Per-sensor temporal model and head. A bidirectional GRU processes each sensor's sequence independently, by reshaping the (B, T, N, d) tensor to $(B{\cdot}N, T, d)$, and a 2-layer MLP maps the final state to the forecast horizon. There is no cross-sensor operation in either. This is also why robustness to sensor dropout is largely built in rather than learned.

The model has $\sim$315K parameters at d=128. Training uses AdamW with cosine annealing and early stopping. We use the Peaks-Over-Threshold loss of [1] throughout, so the controlled comparison in Section IV-C differs only in the spatial backbone.

B. Coverage-Guaranteed Intervals at Unseen Sensors

We wrap OmniST in split conformal prediction [5], [6]. With absolute residuals $s_i = |\hat{x}_i - x_i|$ on a calibration set $\mathcal{C}$ of size n, the half-width is

$$\hat{q} = \text{Quantile}\Big(\{s_i\}_{i \in \mathcal{C}},\ \tfrac{\lceil (n+1)(1-\alpha) \rceil}{n}\Big), \tag{3}$$

and the interval is $[\hat{x} - \hat{q},\ \hat{x} + \hat{q}]$, clipped at zero for non-negative quantities. Under exchangeability this gives marginal coverage of at least $1 - \alpha$.

In the inductive setting the calibration residuals come from training sensors while the test point is an unseen sensor, so the calibration and test scores must remain exchangeable, the issue analyzed for graph models in [3]. For message-passing GNNs this is delicate: adding or removing any node perturbs every embedding and shifts all calibration scores. OmniST's per-sensor construction gives an exact property instead.

Proposition 1 (Locality implies exact score invariance): Let $s_i = g\big(c_i, x_i, \{(c_j, x_j) : j \in \mathcal{N}_k(i; S)\}, y_i\big)$ be the nonconformity score of sensor i given the present sensor set S, where $\mathcal{N}_k(i; S)$ is the set of its k nearest other sensors in S. For any S' with $\mathcal{N}_k(i; S') = \mathcal{N}_k(i; S)$, in particular any addition or removal of sensors outside that set, $s_i(S') = s_i(S)$ exactly. Hence the threshold $\hat{q}$ calibrated on training sensors remains valid without recalibration for any test sensor whose neighbor set is unchanged.

Proof sketch. Each component is a function of a sensor's own data and its k-NN receptive field: the coordinate token depends on (c_i, x_i), the attention aggregates exactly $\mathcal{N}_k(i; S)$, and the GRU and head act per sensor. No operation uses N, a sensor index, or a reduction across sensors, so s_i is a fixed function of an unchanged argument. Validity of $\hat{q}$ then follows from standard split conformal, since neither the calibration scores nor the test score move. □

Empirically, dropping sensors outside a test sensor's neighbor set changes its prediction by less than 10^{-7}, while dropping a true neighbor changes it by about 7×10^{-3}, so the locality is real rather than vacuous.

C. Coverage Over Time

Split conformal is marginal under exchangeability. Over a long horizon the residual distribution drifts with the seasons, and a quantile frozen on an early window under-covers later. We therefore use a rolling-window quantile, which tracks the drifting residual scale, with a multiplicative correction from Adaptive Conformal Inference (ACI) [4],

$$\alpha_{t+1} = \alpha_t + \gamma\,(\alpha - \text{err}_t), \tag{4}$$

where $\text{err}_t = 1$ if the observation fell outside the interval at step t. ACI drives long-run miscoverage to α for any sequence, without exchangeability. We call the combination rolling-ACI. Its cost is wider intervals, which is unavoidable when the residuals widen.

III. Data and Protocol

We use four real sensor datasets from three physical domains:

- **Delhi air quality:** daily $PM_{2.5}$ from 37 CPCB stations (2019–2023), from the E-STGCN public repository [1], the dataset behind its published results. This is our primary accuracy benchmark: training on 2019–2022 (1461 days), testing on 2023 (365 days), with 60-day inputs and a 30-day horizon.
- **Beijing air quality:** hourly $PM_{2.5}$ from the 12-station UCI Multi-Site dataset [7] (2013–2017), aggregated to daily.
- **Western-US weather:** daily maximum temperature from 30 NOAA GHCN-Daily stations (2018–2022) with native coordinates.

- **Freeway traffic:** 5-minute speeds from the 228-sensor Caltrans PeMSD7-M benchmark [2], plus the 325-sensor PeMS-BAY benchmark [8] for scale. Both release a distance adjacency rather than coordinates, so we reconstruct 2D sensor positions by classical multidimensional scaling; the speed series are the original measurements.

For inductive evaluation we hold out geographically spread sensors from training entirely (5 on Delhi), calibrate on training-sensor residuals, and measure coverage and interval width at the held-out sensors. Failure is simulated by removing sensors at inference. Seed counts are stated with each result.

IV. Results

A. Accuracy on the Trained Network

On Delhi, a 5-seed held-out evaluation gives OmniST an MAE of 28.2 ± 0.9 μg/m^3, statistically indistinguishable from E-STGCN on the same split. A month-by-month comparison against E-STGCN's published numbers is given in the supplementary material; because those numbers come from a protocol we did not run, the controlled comparison below is the one we rely on.

B. Sensor Failure

TABLE I
FORECAST MAE UNDER SIMULATED SENSOR FAILURE (DELHI)

Active stations	MAE (μg/m^3)	Degradation
37 (full network)	35.33	—
32 (5 failures)	35.72	+1.1%
27 (10 failures)	35.72	+1.1%
22 (15 failures)	35.58	+0.7%

With up to 15 stations (40% of the network) removed at inference, forecasts at the remaining stations degrade by at most 1.1% (Table I), with no retraining or reconfiguration.

C. Controlled Comparison With a Fixed-Topology Baseline

We reimplemented the E-STGCN architecture family (Chebyshev-GCN spatial module, LSTM temporal module and POT loss [1]) and ran it under identical data, seeds, preprocessing, loss and training budget. This is our reimplementation, not the authors' code (Table II).

TABLE II
OMNIST VERSUS FIXED-TOPOLOGY E-STGCN (5 SEEDS, IDENTICAL CONDITIONS)

Setting	Dataset	OmniST	E-STGCN
Trained (MAE)	Delhi AQ	**28.6**$\pm$1.2	33.5$\pm$1.9
	Weather	5.55$\pm$0.14	**5.30**$\pm$0.29
Unseen	Delhi AQ	MAE 26.7, cov 0.91	*cannot predict*
	Weather	MAE 5.0, cov 0.92	*cannot predict*

On trained sensors OmniST is better on Delhi (paired p=0.03) and E-STGCN is marginally better on weather (p=0.14, not significant). The fixed 32×32 / 27×27 training adjacency has no entry for an unseen sensor, so E-STGCN produces no output there.

On the sensors both models were trained on, neither dominates. The difference appears when the network changes. At an added sensor the baseline has nothing to output. For the mirror case we pin the baseline's adjacency to its training sensors and drop 30% or 50% of them at inference (3 seeds, Delhi and weather): the fixed 32×32 adjacency cannot multiply the 16-sensor tensor that remains, and the run stops with a shape error. OmniST, on the same reduced network and recalibrated on the survivors, holds coverage at the unseen stations ($0.914/0.913$ on Delhi and $0.925/0.927$ on weather under 30%/50% dropout). Both failures share one cause: the baseline binds its weights and adjacency to the training sensor set.

D. Forecasts and Intervals at Unseen Sensors

TABLE III
POINT FORECASTS AT UNSEEN DELHI STATIONS (5 HELD OUT, 3 SEEDS)

Method	MAE (μg/m^3)
Nearest neighbor (copy closest training station)	30.3
IGNNK (learned inductive kriging [9])	34.0$\pm$1.3
OmniST + node-dropout training augmentation	27.1$\pm$1.1
OmniST (coordinate-based)	**26.7**$\pm$1.2

At held-out stations OmniST improves on nearest-neighbor interpolation by about 12% (26.7 against 30.3) and on IGNNK [9], a diffusion-GCN kriging model trained with random subgraph masking (our reimplementation), by about 21% (Table III). The advantage also holds on weather, where OmniST beats IGNNK on inductive MAE (4.99 against 5.92). Node-dropout augmentation adds nothing (27.1, within noise), so the gain comes from the coordinate-based design.

TABLE IV
INTERVALS AT UNSEEN DELHI STATIONS (TARGET 90%, 3 SEEDS)

Method	Coverage	Mean width	Guarantee
Nearest neighbor + residual	0.925	149.7	No
OmniST + conformal	**0.915**$\pm$0.006	**128.5**$\pm$3.7	Yes
30% stations dropped	0.914	—	Yes
50% stations dropped	0.913	—	Yes

A 10-seed run (Table V) gives a consistent 0.914 (width 127.0).

The point forecast matters less than the interval around it. OmniST's intervals reach 91.5% coverage at stations never seen in training and are 14% narrower than the nearest-neighbor baseline (128.5 against 149.7), which over-covers and carries no finite-sample guarantee (Table IV). IGNNK under-covers (0.89 at the 0.90 target), while OmniST holds the target with a 19% narrower interval. Coverage stays near 91% after 30% or 50% of the training stations are removed and the model is recalibrated on the rest.

E. Across Domains, Baselines and Scale

Coverage at unseen sensors stays at or above 90% on all four datasets and holds under 30 to 50% dropout (Table V, Fig. 1). The nearest-neighbor baseline under-covers on three of

TABLE V
COVERAGE AT UNSEEN SENSORS ACROSS FOUR REAL DATASETS FROM THREE DOMAINS (TARGET 90%)

Dataset	OmniST	NN	Drop 30%	Drop 50%
Delhi AQ	**0.914**±.006	0.925	0.912	0.916
Beijing AQ	**0.928**±.003	0.869	0.928	0.930
Weather	**0.919**±.007	0.896	0.919	0.920
Traffic	**0.915**±.008	0.887	0.916	0.917

NN = nearest-neighbor residual baseline. Traffic is PeMSD7-M. 10 seeds (air quality, weather); 6 seeds (traffic).

TABLE VI
OMNIST VERSUS MODEL-FREE INTERPOLATORS, ALL WITH SPLIT CONFORMAL (COVERAGE AT TARGET 90%)

Dataset	OmniST	NN	IDW	Mean-k	GP
Delhi AQ	**0.914**	0.925	0.871	0.813	0.845
Beijing AQ	**0.928**	0.869	0.818	0.789	0.789
Weather	**0.919**	0.896	0.927	0.945	0.893
Traffic	**0.911**	0.887	0.905	0.906	0.922

The traffic value for OmniST (0.911) is from this run; Table V reports 0.915 at 6 seeds. The difference is seed variation.

them (Beijing 0.869, weather 0.896, traffic 0.887). Against four model-free interpolators (nearest neighbor, inverse-distance weighting, mean of k nearest, and a Gaussian-kernel kriging-style predictor), each wrapped in the same conformal procedure, OmniST is the only method at or above target on every dataset (Table VI); all four fall to 0.79–0.87 on Beijing. The interpolators are often tighter, and sometimes better on point MAE, because they under-cover. Point MAE is mixed: OmniST wins on Delhi, and interpolation wins on the denser networks. Ablations in the supplementary material show the coverage comes from the coordinate-based inductive construction rather than from the attention layer or a particular k. The guarantee also tracks other targets: at 80, 90 and 95% (5 seeds for air quality, 3 for traffic), empirical coverage is within about 3 points of the target in every cell, for example Delhi $0.824/0.915/0.958$; the loosest cell is traffic at 80% (0.773).

At 228 traffic sensors inductive coverage is 91.9%. On the full 325-sensor PeMS-BAY network (3 seeds, scalable k-NN) it is $92.1 \pm 0.0\%$ and stays at or above 92.0% under 30/50% dropout, while the nearest-neighbor baseline reaches 87.0%. A scalable neighbor index removes the $O(N^2)$ cost of exact attention; timing results are in the supplementary material.

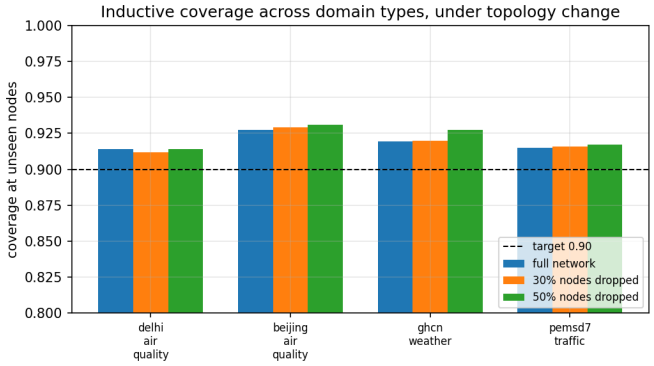


Fig. 1. Coverage at unseen sensors on four real datasets, at full network and under 30% and 50% sensor dropout. OmniST holds the 90% target throughout.

F. Coverage Under Seasonal Drift

TABLE VII
COVERAGE OVER TIME UNDER SEASONAL DRIFT (DELHI, TARGET 90%, 3 SEEDS)

Method	Overall	Late window	Width
Static split conformal	0.758	0.498	95.4
Rolling window	0.840	0.799	127.4
ACI	0.847	0.905	139.0
Rolling-ACI	**0.890**	**0.962**	170.1

Late window = final third of the test year. Width in μg/m^3.

The spatial guarantee does not carry over to time for free. Calibrating on the first half of the test year and evaluating on the second, static conformal covers 0.76 overall and only 0.50 in the final third of the year. Rolling-ACI restores 0.89 overall and 0.96 in the late window, at the cost of wider intervals; ACI alone is a narrower middle ground (Table VII, Fig. 2).

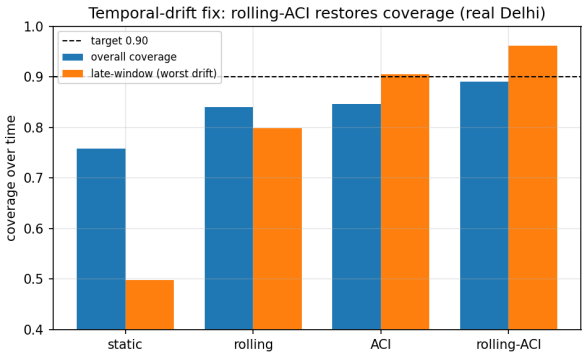


Fig. 2. Coverage over time under seasonal drift on Delhi. Static conformal falls to 0.50 in the late window; rolling-ACI restores the 90% target.

V. RELATED WORK

Dynamic and inductive forecasting. Fixed-topology STGCNs [2] couple weights to a sensor count. Expanding-node methods relax this for sensor addition, and inductive kriging (IGNNK [9]) predicts at unobserved locations; we use it as a baseline. Closest to our architecture, STFO [10] uses coordinate-based aggregation without sensor-specific parameters to absorb network expansion, and cold-start forecasting at unseen sensors [11] conditions a meta-graph on local support. Both target point prediction under growing networks. OmniST handles removal as well as addition, needs no prompting or continual-learning machinery, and attaches a coverage guarantee to its inductive forecasts. Cross-domain spatiotemporal pretraining [12] is a different axis that we do not pursue.

Conformal prediction on graphs and time series. Split conformal [5], [6] has been extended to correlated time series [13], graph time series [14], and energy graphs [15],

and [3] analyzes the inductive setting in which new nodes perturb calibration scores. Recent work gives valid conformal prediction for dynamic GNNs [16], non-exchangeable conformal prediction for temporal GNNs [17], topology-aware intervals for stream networks under shift [18], and online conformalized Gaussian processes on graphs [19]. These mostly assume a fixed node set whose edges or labels evolve. OmniST targets a different case: coverage at sensor coordinates with no training history, maintained as sensors are added or removed. Our online layer follows the spirit of [18], [19]; the novelty is its composition with coordinate-based inductive prediction. Model-free valid spatial prediction [20] motivates our interpolation baselines.

VI. Limitations

- **Marginal, not conditional, coverage.** The guarantee holds on average. Per-sensor coverage on the irregular traffic network ranges from 0.73 to 0.98; Mondrian (group-conditional) conformal lifts the worst group from 0.80 to 0.86 but does not reach 0.90 (supplementary material). Exchangeability of OmniST's inductive scores is argued structurally and checked empirically, not proved in general.
- **Accuracy benchmark.** The comparison with E-STGCN is on Delhi $PM_{2.5}$ only and uses our reimplementation. Coverage is validated on all four datasets.
- **Scale.** Coverage is certified up to 325 real sensors. Thousands of sensors await a larger real benchmark; synthetic node duplication cannot certify coverage because it breaks exchangeability.
- **Coordinates.** For the two traffic datasets, sensor positions are reconstructed from the released distance adjacency by multidimensional scaling.
- **Drift.** Rolling-ACI restores long-run coverage at the price of wider intervals (170 against 95 $\mu g/m^3$ on Delhi); its guarantee is asymptotic and marginal, not per step.
- **Spatial attention.** Its effect on accuracy at trained sensors is below 0.2 $\mu g/m^3$; its value is enabling forecasts and intervals at unseen sensors.
- **Seeds.** Cross-domain coverage results use 10 seeds (6 for traffic); the temporal, inductive-interval and conditional-coverage results use 3.

VII. Conclusion

OmniST is a forecaster for sensor networks that change. Because no component references a global sensor count, it keeps forecasting when 40% of stations fail, losing at most 1.1% accuracy, and it forecasts at sensors it never trained on. Wrapped in inductive split conformal, it gives intervals at those unseen sensors whose coverage holds at or above 90% across four real datasets from three domains, under 30 to 50% dropout, and at 325 real sensors, where model-free interpolators fall short and a fixed-topology baseline cannot produce an output. A rolling adaptive-conformal layer extends the guarantee over time. The open problems are conditional coverage per sensor and per horizon, and certifying coverage on real networks with thousands of sensors.

Data and Code Availability

All datasets are public: Delhi CPCB $PM_{2.5}$ from the E-STGCN repository [1]; Beijing Multi-Site Air Quality from the UCI repository [7]; western-US daily temperature from NOAA GHCN-Daily; PeMSD7-M traffic from [2]; and PeMS-BAY traffic from [8]. The implementation, including the model, loaders for every dataset, each experiment script, a pinned environment, and an index mapping each reported number to its result file, is available at `https://github.com/Praveg432/OmniST`.

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